

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

(192511121)

I T. Subahan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Subahan

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	2	1
Identification of Key Issues	20%	1	1	2	1	1
Application of Concepts	25%	1	1	1	1	1
Decision Making	20%	2	1	1	1	1
Clarity of Response	15%	1	1	1	1	1
		6	5	6	6	5

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Smart Campus Subnet Allocation:

Basic Network Block : 192.168.100.0/24

Subnet Allocations :

- i) School of computing : /25
- ii) School of Electronics : /26
- iii) School of Mechanical Engineering : /27

i. School of Computing (125) :

- Subnet Mask : 255.255.255.128
- Block Size = 128 IP addresses
- Network Addresses : 192.168.100.0
- IP Range : 192.168.100.0 - 192.168.100.127
- Broadcast Addresses : 192.168.100.127
- Valid Host IP Range : 192.168.100.1 - 192.168.100.126
- Usable Host Addresses = 126 hosts

ii. School of Electronics (126) :

- Subnet Mask : 255.255.255.192
- Block Size = 64 IP addresses
- Network Addresses - 192.168.100.128
- IP Range : 192.168.100.128 - 192.168.100.191
- Broadcast Address : 192.168.100.191
- Valid Host IP Range : 192.168.100.129 - 192.168.100.190
- Usable Host Addresses : 62 hosts.

iii. School of Mechanical Engineering (127)

- Subnet Mask : 255.255.255.224
- Block Size = 32 IP addresses
- network addresses - 192.168.100.192
- IP Range : 192.168.100.192 - 192.168.100.223

Routing Table Lookup:

1. Routing Table Entries:

→ 192.168.1.0/24

→ 192.168.2.0/24

→ 192.168.3.0/24

2. Incoming Packet Destination IP:

192.168.2.45

a) Destination Subnet Determination:

Entry 1 (192.168.1.0/24): Range is 192.168.1.0 - 192.168.1.255

Entry 2 (192.168.2.0/24): Range is 192.168.2.0 - 192.168.2.255

Entry 3 (192.168.3.0/24): Range is 192.168.3.0 - 192.168.3.255

b) Matching Routing Table Entry:

matching entry: 192.168.2.0/24

c) Next-Hop Network Interface:

The packet will be forwarded to the egress interface - all or next-hop router connected to subnet interface

d) Default Route Handling:

→ Default route notation: 0.0.0.0/0

If no default route is configured, the router drops the packet and sends an ICMP destination is unreachable.